

Swamp White Oak

Quercus bicolor

Species background: Native to the eastern and midwestern United States and southeastern Canada, especially in wet bottomlands, floodplains, and low-lying areas near rivers. Turn a leaf over -- the pale, softly fuzzy underside contrasts sharply with the dark green top, a trait most other street oaks don't share.

This packet contains one full lesson for each grade band: K-2, 3-5, 6-8, and 9-12. Each includes a learning objective, standards connection, materials, teacher background, a step-by-step procedure, discussion questions, an assessment, and an extension activity.

Swamp White Oak: Feel swamp white oak's fuzzy leaf underside.

Standards connection: NGSS K-LS1-1; National Core Arts Standards VA:Cr1 (K-2)

Materials: Coloring sheet, crayons.

Teacher background

Turn one of my leaves over -- the underside is soft and a little fuzzy! I like wet soil more than most trees, so you might find me near low, damp spots.

Procedure

- 1 Engage: Ask students to describe something soft they've felt before.
- 2 Explore: Gently turn a real or pictured leaf over and feel the underside.
- 3 Explain: Explain that this fuzzy underside is a special clue for this kind of oak.
- 4 Art: Feel a leaf's underside and draw a fuzzy line where it's soft.
- 5 Evaluate: Student can describe the leaf underside as soft or fuzzy.

Discussion questions

Why might this tree like wet, low spots more than other trees?

Assessment

Correctly describes the fuzzy leaf underside.

Extension

Look for a low, damp spot on your walk where this tree might grow well.

Swamp White Oak: Compare swamp white oak's rounded lobes to red oak's pointed ones.

Standards connection: NGSS 4-LS1-1; National Core Arts Standards VA:Cr2 (3-5)

Materials: Coloring sheet, colored pencils.

Teacher background

This oak likes wet soil more than most street trees -- that's why 'swamp' is in its name. Its leaf lobes are rounded, not sharp and pointy like a red oak's.

Procedure

- 1 Engage: Ask what students think 'swamp' means and why it might be part of this tree's name.
- 2 Explore: Compare this leaf's rounded lobes to a red oak leaf's pointed ones if available.
- 3 Explain: Explain that rounded lobes mean this tree is in the 'white oak' group.
- 4 Art: Draw a leaf from both sides on the same page -- dark green on top, pale and fuzzy on the bottom -- like a leaf with two different outfits.
- 5 Evaluate: Student can describe the leaf lobes as rounded, not pointy.

Discussion questions

Why might a name like 'swamp white oak' tell you something useful about where to find the tree?

Assessment

Correctly identifies rounded lobes as a white oak group trait.

Extension

Find a red oak nearby and compare its pointed lobes directly.

Swamp White Oak: Explain flood tolerance and why cities plant this species in stormwater infrastructure.

Standards connection: NGSS MS-LS1-5; National Core Arts Standards VA:Cr2/Re7 (6-8)

Materials: Coloring sheet, mixed-texture art materials.

Teacher background

Swamp white oak's roots have adaptations for surviving low-oxygen, waterlogged soil, which is why cities increasingly plant it in stormwater and rain-garden tree pits.

Procedure

- 1 Engage: Why would waterlogged soil be a problem for most tree roots?
- 2 Explore: Research how stormwater tree pits and rain gardens work in city streets.
- 3 Explain: Explain why flood-tolerant roots make this species useful for that specific green infrastructure.
- 4 Art: Create a two-sided leaf illustration, front and back, using different textures -- smooth marker on top, stippled dots on the bottom -- and label why the underside might need to be different.
- 5 Evaluate: Write a paragraph explaining why this tree is a good fit for stormwater tree pits specifically.

Discussion questions

What other native plants might be good choices for a rain garden, based on similar flood tolerance?

Assessment

Paragraph correctly connects flood tolerance to stormwater infrastructure use.

Extension

Look for a rain garden or stormwater tree pit on your walk and see what's planted there.

Swamp White Oak: Explain root-tissue adaptations for oxygen transport in waterlogged soil.

Standards connection: NGSS HS-LS1-3; National Core Arts Standards Proficient-level Creating/Responding (9-12)

Materials: Coloring sheet, research sources, graphite pencils for a scientific illustration.

Teacher background

Waterlogged, low-oxygen soil kills most tree roots by suffocation; research how swamp white oak's roots develop specialized tissue (aerenchyma) to transport oxygen downward, and explain why this makes it a preferred species for stormwater and rain-garden tree pits in modern green-infrastructure design.

Procedure

- 1 Engage: How could a root possibly get enough oxygen while sitting in standing water?
- 2 Explore: Research aerenchyma tissue and how it transports oxygen down to waterlogged roots.
- 3 Explain: Connect this root-level adaptation to the tree's use in modern green-infrastructure design.
- 4 Art: Sketch the leaf's cross-section as a scientific illustration, showing the fuzzy underside (trichome layer) and labeling it alongside the smooth upper cuticle, referencing real leaf-anatomy diagrams.
- 5 Evaluate: Submit the aerenchyma research summary and the labeled cross-section illustration.

Discussion questions

What other flood-tolerant plant adaptations (besides aerenchyma) might be worth researching for future green-infrastructure plant selection?

Assessment

Rubric covering accuracy of the root-adaptation research and illustration quality.

Extension

Compare swamp white oak's flood-tolerance strategy to a completely dry-adapted species like honeylocust, and discuss why city tree-pit conditions vary so much block to block.

SOURCES & METHODOLOGY

Background research drawn from TreeWalk NYC's own species research. Lesson structure (objective, background, materials, procedure, assessment, extension) follows the format used by real, freely published environmental-education lessons, including the National Park Service's "Looking at Leaves" 5E lesson plan (Acadia Outdoor Classroom Collaborative), Penn State Extension's "Sorting and Classifying Leaves," and Project Learning Tree's leaf-activity guidance -- adapted and written fresh for this species, not copied from those sources. This is an educational pilot; standards references indicate topic alignment, not formal certification.